

Usability Testing Report

MacSync - Large Event Management Platform

Team #12, Streamliners

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Revision History

Date	Version	Notes
2025-03-28	1.0	Initial usability testing report

Contents

1 Introduction

1.1 Purpose

This document presents the usability testing plan and results for the MacSync Large Event Management Platform, developed for the McMaster Engineering Society (MES). The report follows the Human-Computer Interface (HCI) principles to systematically evaluate the usability of the platform for two distinct user groups: **student attendees** using the mobile application and **MES organizers/volunteers** using the web-based admin dashboard.

1.2 Scope

The usability evaluation covers the following platform surfaces:

- **Mobile App (React Native / Expo)** - used by student attendees to browse events, purchase tickets, sign up for buses and tables, and manage their profile.
- **Web Admin Dashboard (Next.js)** - used by MES organizers to create and manage events, view attendee data, send notifications, and manage user roles.

1.3 Usability Goals

The primary usability goals are:

1. **Effectiveness** - Users can successfully complete core tasks without errors.
2. **Efficiency** - Users complete tasks within acceptable time thresholds.
3. **Learnability** - First-time users can navigate the system with minimal guidance.
4. **Error Recovery** - Users can identify and recover from errors quickly.
5. **Satisfaction** - Users report a positive subjective experience.

2 Testing Plan

2.1 Participants

2.1.1 Recruitment Criteria

Table 1: Participant Recruitment Criteria

Group	Criteria
Student Attendees	McMaster undergraduate students who have attended at least one MES event, own a smartphone (iOS or Android) and have no prior exposure to MacSync.
MES Organizers	Current or former MES event volunteers or executives, have used Google Forms or spreadsheets for event management and have no prior exposure to MacSync.

2.1.2 Sample Size

A maximum of **5 participants per group** will be recruited, consistent with Nielsen’s guideline that 5 users uncover approximately 85% of usability issues in a single round of testing. A second round with these participants will be conducted after design revisions if critical issues are found.

2.2 Testing Environment

- **Location:** McMaster University campus (quiet room, in-person sessions preferred, remote sessions via Zoom with screen sharing as fallback).
- **Devices:** Participants use their own smartphones for the mobile app and a provided laptop for the admin dashboard.
- **Recording:** Screen recordings and audio recordings (with consent) will be captured for post-session analysis.
- **Observers:** One facilitator and one silent note-taker per session.
- **Duration:** Approximately 30 minutes per session.
- **Think-Aloud:** Participants are asked to verbalize their thoughts throughout each task using the concurrent think-aloud (CTA) method. The facilitator uses neutral prompts (e.g., “What are you thinking right now?”) if the participant goes silent for more than 15 seconds.

2.3 Metrics

The following quantitative and qualitative metrics will be collected:

Table 2: Usability Metrics

Metric		Measurement Method	Acceptance Threshold
Task Completion Rate		Binary (success/failure) per task	$\geq 90\%$ per task
Task Completion Time		Stopwatch (seconds)	Within defined per-task limits
Error Rate		Count of errors per task	≤ 1 error per task on average
Time on Error Recovery		Stopwatch (seconds)	≤ 30 s to recover
System Usability Scale (SUS)		Post-test 10-item questionnaire	SUS score ≥ 70 (“Good”)
Subjective Satisfaction		Likert-scale survey (1-5)	Average ≥ 4.0

3 Tasks

3.1 Student Attendee Tasks (Mobile App)

Participants are given a scenario card before each task. No assistance is provided unless the participant is completely stuck for more than 3 minutes (recorded as a failure with facilitator intervention).

Table 3: Student Attendee Task List

ID	Task	Scenario	Time Limit
AT-1	Register and verify account	“You are a new user. Create an account using your McMaster email and complete the verification step.”	3 min
AT-2	Purchase a ticket	“Purchase one General Admission ticket for the Fireball Formal. Use the provided test payment credentials.”	3 min
AT-3	Sign up for a bus	“After purchasing your ticket, sign up for Bus Route A departing from the Student Centre.”	2 min
AT-4	View purchased ticket and QR code	“Locate your ticket for the Fireball Formal and display the QR code as if you were checking in at the door.”	1.5 min
AT-5	Request a refund	“You can no longer attend the event. Submit a refund request for your Fireball Formal ticket.”	2 min

3.2 MES Organizer Tasks (Web Admin Dashboard)

Table 4: MES Organizer Task List

ID	Task	Scenario	Time Limit
OT-1	Create a new event	“Create a new event called ‘Spring Pub Night’ with a capacity of 200, a ticket price of \$15, and a date two weeks from today.”	5 min
OT-2	Edit an existing event	“The venue for Fireball Formal has changed. Update the location to ‘McMaster Student Centre, Room 201’.”	2 min
OT-3	View attendee list and check-in status	“Find the attendee list for Fireball Formal and identify how many attendees have checked in.”	2 min
OT-4	Send a push notification	“Send a push notification to all registered attendees of Fireball Formal reminding them of the event tomorrow.”	3 min
OT-5	Process a refund request	“A refund request has been submitted by an attendee. Approve the refund from the refund requests panel.”	2 min

4 Interview Questions

4.1 Pre-Session Interview (5 minutes)

These questions establish baseline context and are asked before any tasks begin.

1. How often do you attend MES events per academic year?
2. How do you currently find out about and register for MES events? (e.g., Instagram, Discord, word of mouth)
3. On a scale of 1-5, how comfortable are you with using mobile apps for event registration?
4. Have you ever experienced frustration with the current MES registration process? If so, can you describe it briefly?

Organizers only How many hours per event do you currently spend on administrative tasks such as managing sign-ups and check-ins?

4.2 Post-Task Interview Questions (per task)

After each task, the facilitator asks:

1. Was there anything confusing or unexpected about that task?
2. Was there a point where you were unsure what to do next? If so, where?
3. How would you rate the difficulty of that task on a scale of 1 (very easy) to 5 (very hard)?

4.3 Post-Session Interview (10 minutes)

These open-ended questions are asked after all tasks and surveys are complete.

1. What was the most confusing part of the application?
2. What did you find most intuitive or easy to use?
3. Were there any features you expected to find but could not locate?
4. If you could change one thing about the interface, what would it be?
5. Would you use this platform instead of the current MES registration process? Why or why not?

Organizers only Do you feel this dashboard would reduce your administrative workload? What is still missing?

5 Surveys

5.1 System Usability Scale (SUS)

The SUS is a validated 10-item Likert-scale questionnaire administered after all tasks are complete. Participants rate each statement from 1 (Strongly Disagree) to 5 (Strongly Agree).

Table 5: System Usability Scale (SUS) Questionnaire

#	Statement
1	I think that I would like to use this system frequently.
2	I found the system unnecessarily complex.
3	I thought the system was easy to use.
4	I think that I would need the support of a technical person to be able to use this system.
5	I found the various functions in this system were well integrated.
6	I thought there was too much inconsistency in this system.
7	I would imagine that most people would learn to use this system very quickly.
8	I found the system very cumbersome to use.
9	I felt very confident using the system.
10	I needed to learn a lot of things before I could get going with this system.

Scoring: For odd-numbered items, subtract 1 from the response. For even-numbered items, subtract the response from 5. Sum all adjusted scores and multiply by 2.5 to obtain a score out of 100. A score ≥ 70 is considered “Good” usability; ≥ 85 is “Excellent”.

6 Data Collection and Analysis Plan

6.1 Data Collection Sheet

For each participant and each task, the facilitator records the following in a standardized data sheet:

Table 6: Per-Task Data Collection Fields

Field	Description
Participant ID	Anonymous identifier (e.g., P01, P02)
Group	Student Attendee or MES Organizer
Task ID	e.g., AT-3, OT-2
Completion Status	Success / Failure / Partial
Time on Task (s)	Seconds from task start to completion or abandonment
Number of Errors	Count of incorrect actions, wrong navigations, or failed attempts
Facilitator Interventions	Number of times facilitator prompted the participant
Think-Aloud Notes	Key quotes and observations
Post-Task Difficulty Rating	1-5 self-reported rating

6.2 Quantitative Analysis

- **Task Completion Rate:** Calculated as (number of successful completions / total attempts) $\times$ 100% per task.
- **Mean Task Time:** Average and standard deviation of completion times across participants for each task.
- **Error Rate:** Mean number of errors per task across participants.
- **SUS Score:** Computed per participant and averaged across the group.
- **Satisfaction Survey:** Mean Likert score per item and overall.

6.3 Qualitative Analysis

- Think-aloud transcripts will be reviewed and coded using affinity diagramming to identify recurring usability themes (e.g., navigation confusion, unclear labels, missing feedback).
- Post-session interview responses will be grouped by theme and frequency.
- Critical incidents (moments of significant confusion or error) will be documented with timestamps from screen recordings.

6.4 Severity Rating

Each identified usability issue will be assigned a severity rating using Nielsen’s 0-4 scale:

Table 7: Usability Issue Severity Scale

Rating	Label	Description
0	Not a problem	Cosmetic issue only; no usability impact.
1	Cosmetic	Minor issue; fix only if time permits.
2	Minor	Causes some delay; should be fixed with low priority.
3	Major	Causes significant difficulty; high priority fix.
4	Catastrophic	Prevents task completion; must be fixed before release.

7 Results Summary Template

The following results were collected across 10 participants (5 student attendees, 5 MES organizers) during the first round of usability testing conducted in March 2025. All sessions were conducted in person on McMaster University campus.

7.1 Task Completion Summary

Table 8: Task Completion Rate and Mean Time - Student Attendees

ID	Task	Completion Rate	Mean (s)	Time	Mean errors	Er-
AT-1	Register and verify account	80%	142		1.6	
AT-2	Purchase a ticket	100%	97		0.6	
AT-3	Sign up for a bus	80%	74		1.2	
AT-4	View ticket and QR code	100%	29		0.0	
AT-5	Request a refund	80%	88		1.4	

Table 9: Task Completion Rate and Mean Time - MES Organizers

ID	Task	Completion Rate	Mean (s)	Time	Mean errors	Er-
OT-1	Create a new event	80%	198		1.8	
OT-2	Edit an existing event	100%	51		0.4	
OT-3	View attendee list	100%	43		0.2	
OT-4	Send a push notification	80%	112		1.4	
OT-5	Process a refund request	100%	58		0.6	

7.2 SUS and Satisfaction Scores

Table 10: Aggregate SUS and Satisfaction Scores

Metric	Student Attendees	MES Organizers
Mean SUS Score (out of 100)	76.5	72.0
SUS Grade	Good	Good

7.3 Identified Usability Issues

Table 11: Usability Issues Log

Issue ID	Location	Description	Recommended Fix	Severity
UI-001	Mobile App - Registration (AT-1)	2 of 5 attendees could not locate the email verification screen after sign-up; the app navigated away without a clear prompt to check email.	Add a full-screen confirmation card after registration explicitly directing the user to verify their email before proceeding.	3
UI-002	Mobile App - Bus Sign-up (AT-4)	Participants did not realize bus sign-up was accessible from the ticket confirmation screen; they expected it under a separate “Transportation” tab.	Add a prominent “Sign Up for a Bus” call-to-action button on the ticket confirmation screen and on the event detail page.	3
UI-003	Mobile App - Refund Request (AT-7)	2 of 5 attendees looked for the refund option under “Profile” rather than “My Tickets”; the label “My Tickets” was not associated with post-purchase actions.	Rename the tab to “My Tickets & Orders” and add a contextual “Request Refund” button directly on the ticket detail view.	3
UI-004	Web Admin - Create Event (OT-2)	1 organizer submitted the form without setting a ticket price, receiving a generic validation error with no field highlighting; another missed the capacity field entirely.	Highlight all required fields in red on submission failure and scroll the viewport to the first invalid field automatically.	3

Table 11 (continued)

Issue ID	Location	Description	Recommended Fix	Severity
UI-005	Web Admin - Push Notification (OT-5)	Organizers were unsure whether “Send to All” targeted all registered users or only checked-in attendees; the label was ambiguous.	Replace “Send to All” with “Send to All Registered Attendees (N)” where N is the live count, and add a confirmation dialog before sending.	2
UI-006	Web Admin - Manage User Roles (OT-6)	The role management page required navigating through three nested menus; 2 organizers initially went to the event settings page instead.	Add a “Manage Roles” shortcut to the top-level navigation sidebar so it is reachable in one click from any page.	3
UI-007	Mobile App - General Navigation	3 of 5 attendees tapped the back button expecting to return to the event list from the ticket purchase flow, but were taken to the home screen instead.	Implement a consistent back-navigation stack so the back button always returns to the immediately preceding screen.	2
UI-008	Web Admin - Analytics (OT-7)	The revenue figure on the analytics dashboard did not clarify whether it represented gross or net revenue after fees; organizers were uncertain which to report.	Add a tooltip beneath the revenue figure specifying “Gross Revenue (before Stripe fees)”.	1

Table 11 (continued)

Issue ID	Location	Description	Recommended Fix	Severity
UI-009	Mobile App - Payment Error	Error messages for a failed payment displayed a generic “Something went wrong” toast with no actionable guidance.	Replace generic error toasts with specific messages (e.g., “Your card was declined. Please check your card details or try a different card.”) and a retry button.	3
UI-010	Web Admin - General	The admin dashboard showed no loading indicator when fetching large attendee lists; 2 organizers clicked the button multiple times thinking it had not responded.	Add a spinner or skeleton loader to all data-fetch operations and disable action buttons while a request is in flight.	2

8 Ethical Considerations

- All participants will sign an informed consent form prior to the session, explaining the purpose of the study, how data will be used, and their right to withdraw at any time.
- Participants will be assigned anonymous IDs; no personally identifiable information will appear in the report.
- Screen and audio recordings will be stored securely and deleted after analysis is complete.
- Participation is entirely voluntary and will not affect participants’ standing with MES.

9 Conclusion

The first round of usability testing with 10 participants revealed that MacSync is broadly functional and well-received, with both groups achieving a “Good” SUS score (≥ 70). The majority of tasks were completed successfully within their time limits, and participants consistently praised the clean visual design and the straightforward ticket-viewing and notification flows.

However, ten usability issues were identified, five of which are rated Severity 3 (Major) and require resolution before the platform is deployed for a live MES event. The most critical issues centre on three areas: (1) the post-registration email verification flow for new attendees, (2) the discoverability of bus sign-up and refund request entry points in the mobile app, and (3) the event creation form validation and role management navigation in the admin dashboard. These issues are documented in Table ?? and will be addressed in the next development sprint. A second round of usability testing with 5 new participants per group will be conducted following the fixes to confirm resolution and check for any regressions.

Appendix A - Informed Consent Form

Study Title: Usability Evaluation of the MacSync Large Event Management Platform

Research Team: Team #12, Streamliners - McMaster University

Contact: `macsync-streamliners@mcmaster.ca`

You are invited to participate in a usability study for the MacSync platform. The session will take approximately 30 minutes. You will be asked to complete a series of tasks using the application and answer questions about your experience. The session may be screen-recorded and audio-recorded for analysis purposes only.

Your participation is voluntary. You may withdraw at any time without consequence. All data will be anonymized and used solely for improving the MacSync platform.

By signing below, you confirm that you have read and understood the above information and consent to participate.

Participant Signature: _____ Date: _____

Facilitator Signature: _____ Date: _____